Goursat's Theorem

Theorem 0.0.3. Let G be an open set and let $f : G \to \mathbb{C}$ be a holomorphic function.
Then,

$$\int_T f = 0$$

for every triangular path T in G.

Proof. Let $T_0 = [a, b, c, a]$ be the triangular path in G, and let $d = \operatorname{diam} T$ and $L = \operatorname{length} T$.
And, construct four subtriangles

$$T^{(1)} = \left[a, \frac{a+b}{2}, \frac{a+c}{2}, a\right],\ T^{(2)} = \left[\frac{a+b}{2}, b, \frac{b+c}{2}, \frac{a+b}{2}\right],\ T^{(3)} = \left[\frac{a+c}{2}, \frac{b+c}{2}, c, \frac{a+c}{2}\right],\ T^{(4)} = \left[\frac{a+b}{2}, \frac{b+c}{2}, \frac{a+c}{2}, \frac{a+b}{2}\right]$$

Then,

$$\int_{T_0} f = \int_{T_1} f + \int_{T_2} f + \int_{T_3} f + \int_{T_4} f$$

Thus, by the triangle inequailty,

$$\left|\int_{T_0} f\right| \leq \left|\int_{T_1} f + \int_{T_2} f + \int_{T_3} f + \int_{T_4} f\right| \leq 4 \cdot \max_{i=1,2,3,4} \left|\int_{T_i} f\right|$$

Let $T_1 = T^j$, where the j maximizes the above integral. Apply the same process for $T_0, T_1, \ldots, T_{n-1}$, we obtain that a sequence of nested compact sets $\{\mathcal{T}_n\}$, where $\mathcal{T}_n$ is the convex hull of T_n. So, the intersection

$$\bigcap_{n=0}^{\infty} \mathcal{T}_n = \{z_0\}$$

is a singleton. Given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|z - z_0| < \delta \implies \left|\frac{f(z) - f(z_0)}{z - z_0} - f'(z_0)\right| < \varepsilon$$

since f is holomorphic. Moreover, by the construction, there exists $N \in \mathbb{N}$ such that

$$T_N \subseteq B(z_0; \delta)$$

Now, the triangle inequality again,

$$\left|\int_{T_N} f(z)dz\right| = \left|\int_{T_N} f(z) - f(z_0) - f'(z_0)(z - z_0)dz\right| \leq \varepsilon \int_{T_N} |z - z_0||dz| = \varepsilon \cdot \operatorname{diam} T_N \cdot \operatorname{length} T_N$$

Finally, since $\operatorname{diam} T_N = 2^{-n} \operatorname{diam} T_0$ and $\operatorname{length} T_N = 2^{-n} \operatorname{length} T_0$,

$$\left|\int_{T_0} f(z)dz\right| \leq 4^N \cdot \left|\int_{T_N} f(z)dz\right| \leq 4^N \cdot \varepsilon \cdot \operatorname{diam} T_N \cdot \operatorname{length} T_N \leq \varepsilon \cdot \operatorname{diam} T_0 \cdot \operatorname{length} T_0 \to 0 \ \text{ as } \ \varepsilon \to 0$$

□

[Thm. Let G be an open set of $\mathbb{C}$, and T be a triangular path contained in G.
For a point a inside in T, suppose that f is Holomorphic on $G \setminus \{a\}$.
If f is bounded on for some neighborhood of a, then

$$\int_T f = 0,$$

Proof. For fixed a, construct a subtriangle T^1, T^2, T^3, T^4 from T
Then, for some T^k, $a \in T^k$. (There are at most three cases).
Put this $T_1 = T^k$. Then, inductively, we can construct $\{T_n\}$, compact nested, and
$\left|\int_T f\right| \le \left|\sum_{a \in T_n^k} \int_{T_n^k} f\right| \le \frac{3L}{2^n} \cdot \sup|f| \to 0$ as $n \to 0$.

[Morera Theorem.
Let $f : D \to \mathbb{C}$ be a continuous function on an open set D.
Suppose that for every triangle T inside G,

$$\int_T f = 0.$$

Then, f is Holomorphic on G.
Proof. Let $z_0 \in G$ be fixed, and $B_r(z_0)$ be given where r was chosen by $B_r(z_0) \subseteq G$. For each $z \in B_r(z_0)$, define

$$F(z) = \int_{[z_0, z]} f(u)\, du \quad \text{where } [z_0, z] \text{ is the straight segment.}$$

For small $h \in \mathbb{C}$, observe that

$$\frac{F(z+h) - F(z)}{h} = \frac{1}{h} \cdot \int_{[z, z+h]} f(u)\, du, \text{ because}$$

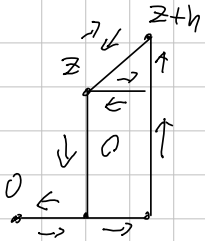
$$\int_{[z_0, z]} f + \int_{[z, z+h]} f + \int_{[z+h, z_0]} f = 0, \text{ by the assumption.}$$

Now, the continuity gives $\lim_{h \to 0} \int_{[z, z+h]} f(u)\, du = f(z)$. □

Thm. A holomorphic function $f: D \to \mathbb{C}$ has a primitive where D is open disk.

Pf. WLOG, let $D = B(0; R)$. Given $z = a + bi \in D$, define

$$F(z) = \int_{\gamma_z} f(s)\,ds \quad \text{where } \gamma_z = [(0,0), (a,0), (a,b)]$$



Now, let $h \in \mathbb{C}$ with $0 < |h| < R - |z|$. Then,

$$F(z+h) - F(z) = \int_{\gamma_{z+h}} f(s)\,ds - \int_{\gamma_z} f(s)\,ds = \int_{[z, z+h]} f(s)\,ds$$

using Goursat Thm in Here, with holomorphic.

Given $\varepsilon > 0$, there is $\delta > 0$ such that

$$|z - s| < \delta \Rightarrow |f(z) - f(s)| < \varepsilon.$$

So, $\left| \frac{1}{h} \cdot \int_{[z, z+h]} f(s)\,ds - f(z) \right| = \left| \frac{1}{h} \cdot \int_{[z, z+h]} f(s) - f(z)\,ds \right| < \varepsilon$, for small enough h. ∎

Corollary. If f is Homolophic on an open disk, then for any closed curve γ in the disk,

$$\int_\gamma f = 0.$$

Cauchy integral formula

Thm. Let $D=\{z\in\mathbb{C} \mid |z-z_0|\le r\}$ be a disk centered at z_0, and $C: t\mapsto z_0+r\cdot e^{it}$ be a curve, along the boundary of D.
Suppose that $f: G\to\mathbb{C}$ is a Holomorphic on an open G containing D.
Given $z\in D$, $f(z)=\frac{1}{2\pi i}\cdot\int_C \frac{f(w)}{w-z}\,dw$

Proof. Let $z\in D$ be fixed. Since $f(w)/w-z$ is holomorphic except $w=z$, for a cure $C_\varepsilon: t\mapsto z+\varepsilon\cdot e^{it}$ where $\varepsilon<r$, we obtain

$$\int_{-C_\varepsilon}\frac{f(w)}{w-z}\,dw+\int_C\frac{f(w)}{w-z}\,dw=0, \quad \cdots (*)$$

by the Cauchy-Goursat theorem.
Since f is Holomorphic, the derivative of f near z is bounded, so

$$\int_{C_\varepsilon}\frac{f(w)-f(z)}{w-z}\,dw\to 0 \quad \text{as} \quad \varepsilon\to 0.$$

and, for any $\varepsilon>0$,

$$\int_{C_\varepsilon}\frac{f(z)}{w-z}\,dw=f(z)\cdot\int_0^{2\pi}\frac{1}{\varepsilon\cdot e^{it}}\cdot i\cdot\varepsilon e^{it}\,dt=f(z)\cdot 2\pi i.$$

Therefore,

$$\int_{-C_\varepsilon}\frac{f(w)}{w-z}\,dw=-\int_{C_\varepsilon}\frac{f(w)-f(z)}{w-z}\,dw-\int_{C_\varepsilon}\frac{f(z)}{w-z}\,dw\to -2\pi i\cdot f(z),$$

as $\varepsilon\to 0$, so we obtain the result, taking $\varepsilon\to 0$ to $(*)$. ∎

Thm. Suppose f is Holomorphic on an open G containing a closed disk D having center z_0,

$$f(z)=\sum_{n=0}^{\infty}\frac{f^{(n)}(z_0)}{n!}\cdot(z-z_0)^n \qquad (z\in D)$$

Pf. $f(z)=\frac{1}{2\pi i}\cdot\int_C\frac{f(w)}{w-z}\,dw$ $\left(\frac{1}{w-z}=\frac{1}{w-z_0-(z-z_0)}=\frac{1}{w-z_0}\cdot\frac{1}{1-\left(\frac{z-z_0}{w-z_0}\right)}\right)$

$$=\frac{1}{2\pi i}\cdot\int_C f(w)\cdot\sum_{n=0}^{\infty}\frac{1}{(w-z_0)^{n+1}}\cdot(z-z_0)^n\,dw$$

$$=\sum_{n=0}^{\infty}\left(\frac{1}{2\pi i}\cdot\int_C\frac{f(w)}{(w-z_0)^{n+1}}\,dw\right)\cdot(z-z_0)^n$$

$$=\sum_{n=0}^{\infty}\frac{f^{(n)}(z_0)}{n!}\cdot(z-z_0)^n.$$

Corollary, Liouville's Theorem.
If $f:\mathbb{C}\to\mathbb{C}$ is entire and bounded, then it is constant.
Proof. Given $z_0\in\mathbb{C}$ and $R>0$, $|f'(z_0)|\le\frac{1}{R}\cdot\sup_{C_R}|f|\to 0$ as $R\to\infty$. Therefore, $f'=0$.
Thus f is constant.

Fundamental Theorem of Algebra.
Non-constant polynomial $P(z)=a_nz^n+a_{n-1}z^{n-1}+\cdots+a_0\in\mathbb{C}[z]$ has a root in $\mathbb{C}$.
Pf. Suppose that $P(z)$ has no root in $\mathbb{C}$. Then, $1/P(z)$ is well-defined on $\mathbb{C}$, entire.
observe that $\frac{P(z)}{z^n}=a_n+\left(\frac{a_{n-1}}{z}+\cdots+\frac{a_0}{z^n}\right)$ $(z\neq 0)$.

The terms $a_{n-k}\cdot z^{-k}$ $(k=1,\cdots,n)$ converges to zero as $|z|\to\infty$, thus for some $R>0$,

$$|P(z)|\ge 2|a_n|\cdot z^n \qquad (|z|>R)$$
$$\ge 2|a_n|\cdot R^n$$

Therefore it is bounded below in $|z|>R$, and clearly in $|z|\le R$.
Now, $\frac{1}{P}$ is bounded, thus constant. Contradiction.